

Automated Email Sender

ABSTRACT

Manual sending emails to many people again and again is not an easy task. It can consume a lot of our time. Whether it is reminding students about assignment deadline, sending weekly reports to a team, or notifying a group of customers about an update or a product, doing all this by manually by typing everything by hand and sending messages and adding files one at a time and hoping you don't accidentally skip someone or send the wrong attachment to the wrong person is a big task and time consumable. To overcome this, we suggest using an Automated Email Sender.

The problem this project tackles is simple to see, but genuinely it is a big headache for everyone. Manual email communication is slow and leads to human error, especially when it has to happen on a regular basis or recurring basis or when we need to send email to many recipients at once. There wasn't a free, easy to set up way to compose and send those emails automatically without either paying for a heavy marketing tool or writing the same script over and over for every new use case. This project fills that gap with a reusable Python-based system.

To build it, Python was the natural choice because of how well it works with email protocols. The core of the project relies on the SMTPLIB Library. This library is used to handle the actual connection to the mail server along with the secure login and email sending process. Along with that, the email library is used to properly structure each message and attach files like PDF, images or documents without breaking formatting. For automatic scheduling, the schedule library is used, along with Python's time library, to handle date and time models keep track of when email should be sent, whether it is a single scheduled send or something that repeats daily or weekly.

This system itself is split into a few clear parts, each handles one job. One part handles the subject and body of the email, including changing the recipient's name and other details. Another part connects to the SMTP server securely and manages opening and closing that connection safely. The scheduler keeps an eye on the timing so emails go out exactly when they supposed to. The recipient manager reads through the list of people who need to receive the email, looping through each one without mixing up any details.

Putting together the flow is simple. The scheduler starts the process. The recipient list gets loaded. Then each email is created and attachment are added if needed. Next, the system connects to the SMTP server and sends the email. The result is logged before moving to the next recipient. Once this cycle is set up correctly, it can continuously keep running on its own for days or weeks.

By the end of this project, what this project delivers is an easier to use automated mailing system that takes manual email sending and turns it into something that runs quietly in the background, saving time, cutting down on mistakes, and making everyday communication easier.